

Lecture Homework Week 06 - Thursday

Template

Student Information

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Student Responses

Answer the following questions

Question 1

When running the Keyboard Demo in QEMU after introducing the bug, briefly describe how it behaves differently than before. Try a few keystrokes in the QEMU window and describe what you see :

The screen appears to be glitching, and when I input multiple keystrokes, only 1 is displayed. They overwrite each other.

Question 2

What do you think is happening? The program continues to execute, but what happens after it returns from `irq_handler`? What C subroutine gets called over and over?

Because the stack pointer is not at its expected position when the program returns, unexpected behavior happens. In this case, `irq_handler` gets called again. We can see this as only 1 character is displayed on the screen at a time.

Question 3

Look at the terminal where the buggy version is running, what messages are being printed out about the timers? Do the timers work in this buggy version?

In terminal, "Timer with delta zero, disabling" keeps getting repeated. I don't think timers work in this version because of the corruption in the `irq_handler`.